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# HTML

\*] What is HTML :

- .) HTML stands for hypertext Markup language.
- .) It is the standard lang for creating web pages.
- .) It describes the structure of a web page.
- .) It consists of series of elements.
- .) Link other doc into our doc is hyper link like

the hypertext

H } = linking <sup>other</sup> doc into our doc (i.e. cursor)  
T }  
M = Tags < > element. <sup>structure</sup>  
L = Creating website (language which is this)

\*] My first html code :

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title> My website </title>  
  </head>
```

<!DOCTYPE html> // 2-01

<html> // open html tag

<head> // open head tag

<title> my first web page </title>

<body>

<h1> hii my name is PK </h1>

</body> // closing body tag

</head> // closing head tag

</html> // closing html tag

<!DOCTYPE html> it defines this document is an HTML document to the browser.

<html> - open tag

The both tags defines in between lines of code are the html documents.

</html> - close tag

<head> - head of the html tag

<title> - title of the tag.

<body> - defines the documents body and it is a container for all contents i.e. heading, paragraph, image, hyperlinks, tables, lists.

<h1> - element defines a large heading

What is an HTML element?

An HTML element is defined by a start tag, some content and an end tag

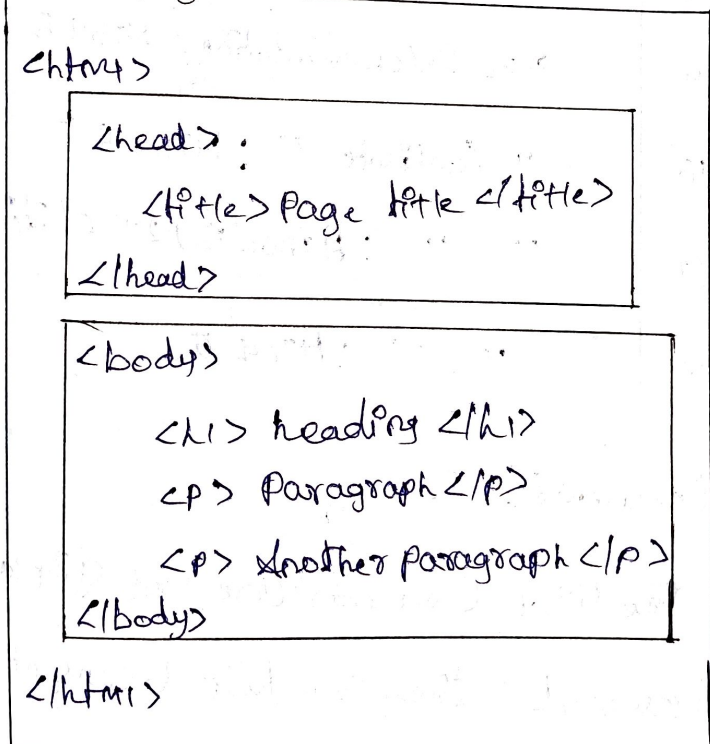
<tagname> Content goes here </tagname>

The HTML element is everything from the start tag to end tag.

| Start tag | Element Content | End tag |
|-----------|-----------------|---------|
| <h1>      | Hi Hi Hi        | </h1>   |
| <p>       | Bye Guys        | </p>    |
| <br>      | None            | </br>   |

Some elements have no content like <br> element. These elements are called empty elements. Empty element do not have an end tag.

### HTML Page Structure:



The Content inside body section will be display on browser. The Content inside title will be display on title bar or on the page tab.

## HTML History :

Since the early days of the world wide web, there have been many versions of HTML.

| Year | Version                           |
|------|-----------------------------------|
| 1989 | Tim-Berners Lee Invented WWW      |
| 1991 | He Invented HTML                  |
| 1993 | Dave Raggett drafted HTML +       |
| 1995 | HTML Working Grp defined HTML 2.0 |
| 1997 | W3C Recommendation : HTML 3.2     |
| 1999 | " " " " HTML 4.0                  |
| 2000 | " " " " XHTML 1.0                 |
| 2008 | WHATWG HTML5 first public draft   |
| 2012 | " " " " Living standard           |
| 2014 | W3C Recommendation : HTML 5       |
| 2016 | " Candidate " : HTML 5            |
| 2017 | " " : HTML 5.1 2nd edition        |
| 2017 | " " : HTML 5.2                    |

## HTML Comments :

The HTML Comments are not displayed in the browser, but they can help document your Source Code.

Comment Tag: `<!--lines-->` (Declaration in start but not in end. (In line Comment)).

Comments are not displayed by the browser, but they can help document your source code.

`<!-- This is a Comment -->`

Hide a line of code:

`<h1> This is a heading </h1>`

`<!-- <h1> This is a heading </h1> -->`

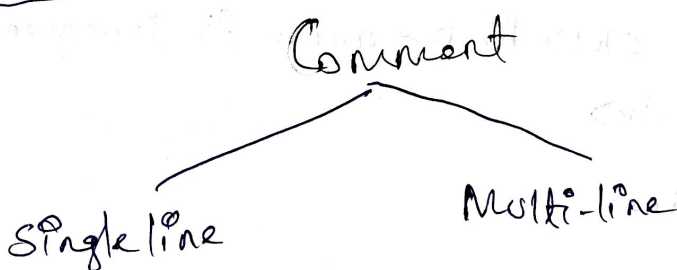
You can also hide more than one line. Everything between the `<!--` and `-->` will be hidden from display.

Hide Inline Comment:

`<p> <!-- heading no para --> This is para </p>`

`<!--` This marks the beginning of HTML Comment.

`-->` This marks the ending of HTML Comment.



## 1) Heading Tag:-

→ The  $h1$  ( $<h1>$  or  $</h1>$ ) is the biggest tag. There are 6 tags the each of the tags are different and slowly decrease the size given on the specific tag.

→ HTML headings are titles or subtitle that you want to display on webpage.

→ It is defined with 6 tags  $<h1>$  defines most imp heading and  $<h6>$  defines least imp heading.

→  $<h1>$  should be used for headings only. ~~Not~~ and  $<h2>$  for headings and less important  $<h3>$ ...

eg:-

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title> my web-site </title>
```

```
</head>
```

```
<body>
```

```
<h1> hi my name is Ironman </h1>
```

```
</body>
```

```
</html>
```

## § Paragraph tag :-

The `html <p>` element defines a paragraph. A para starts on a new line and browsers automatically add some white space before and after a paragraph. (but use `<p>` to add a line after the 1st para).

Solution is

The `html <pre>` element defines preformatted text. The text inside a `<pre>` element is disp in a fixed-width font (usually Courier), and it preserves both spaces and lines breaks.

eg:-

```
<!DOCTYPE html>
```

```
<html>
```

```
  <p> hi my name is Jitesh Kumar  
    i'm 19 years old male </p>
```

```
  <pre> And i also undergone kidney transplant  
last month 9th march 2025 </pre>
```

```
</html>
```

## § The attribute tag :- (imp)

The `<a>` tags defines a hyperlink. The `href` attribute specifies the URL of a page link goes to.

href ->

1. HTML elements have attributes. It provides additional info about elements specified in start tag: (`<a href = " " >Content </a>`). It usually come in pairs like Name = "Value".

Syntax :

`<a href = "www.google.com">` Click here `</a>`  
link                      Attribute (properties of a tag)  
tag

Eg:

```
<!DOCTYPE html>
```

```
<html>
```

```
<a href = "https://maevom-4h.myshopify.com">  
Click here </a>
```

Image tag:

It is the self closing tag. It does not has a closed tag. Use Src (source) Src is also an attribute.

Eg:

```
<!DOCTYPE html>
```

```
<html>
```

```
<img src = "my image.png"> <img src = "my image.png">
```

```
</html>
```

Adjusting Img size.

Use the attributes height and width to reduce or increase the img size.

Syntax: height = "300px" width = "200px"

Eg:

```
<!DOCTYPE html>
```

```
<html>
```

```
<img Src = "myImage.png" height = "300px">
```

```
</html>
```

3) alt text (img is not displayed then the alt text is displayed).

Syntax:-

```
<img Src = "myImage.png" height = "300px" alt = "This  
is an image of goms">
```

4) HR and BR Tags:

HR: This tag draws a horizontal line.

BR: New line (Break tag (BR)).

These two are self closing tags.

Eg:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<Title> my website </Title>
```

```
</head>
```

```
<hr>
```

```
<body>
```

```
<h1> @ hi </h1>
```

```
</body>
```

```
</br>
```

```
</head>
```

my website

hi

5] Absolute URL and Relative URL.  
URL - Universal <sup>Source</sup> Relat<sup>ive</sup> Locator.

Abs url : it points to another website

`<a href = "www.fb.com"> click </a>`

Rel url : it points to a file within a site

`<a href = "myimage.png"> dinesh here </a>`

6] Empty tag:

Empty tags are used when some elements don't need content in between their and closing tags. Sometimes if we does not use any tags its not a problem but continuing this is not recommended.

eg:

`<!DOCTYPE html>`

`<html>`

`<head>`

`<title> hi </title>`

`</head>`

`<body>`

`<h1> im dinesh`

`</body>`

`</html>`

it will print

dis the o/p. but

not continuously

doing this is

not appreciable.

7] Not Case Sensitive :-

Code, tag, body, attributes are coded in small letters.

HTML is ~~case~~<sup>not</sup> case-sensitive

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8] Styling Attribute :-

Style is an attribute. (Colour, Background Colour, font-family, font size, Text align, Border).

This is used to add a style to an element, such as font, size and more.

Syntax :-

<tagname style = "Property : Value ;">

The Property is a CSS property and Value is CSS value.

Syntax for Colour, Bg Colour, font : size, family, Text align, Border).

<!DOCTYPE html>

<html>

<head> <title> HiHello </title>

<body>

<h1 style = "Colour : blue ;">

</h1>

font-family :  
Colour, font size, Text align, ~~Border~~

<h1 style = "Colour : blue ; font-family : verdana ;

~~font-size~~ font-size : 50px ; text-align : centre ;">

background colour

```
<body style="background-color: grey";>
```

border:

```

```

9] Single and double Quotes:-

Title Attribute / Tool tip When you Hover (Cursor Over) on the img then the title attribute can be used.

```
 or - else  
title="'Dinesh Kumar 'DK''">
```

10] bold/strong Tag (formatting elements).

<b> - bold Text

<strong> - Important text.

Syntax:

```
<body>
```

```
<b> hello guys </b>
```

```
<strong> This is really serious matter  
</strong>
```

```
</body>
```

11) I and em tag :

<i> - Italic tag

<em> - Emphasized tag

The HTML <i> tag defines a part of text in alternate voice or mood. The content inside is displayed in Italic. It is often used to indicate a technical term, a phrase from another language, a thought, a ship name etc...

The HTML <em> element defines emphasized text. The content inside is displayed in Italic.

Syntax:

<i> I'm a Billionaire </i>

<em> I'm a Billionaire </em>

12) Small Tag

The HTML <small> element defines smaller text.

Syntax:

<small> This is a smaller text </small>

13) Mark Tag:

Highlighting the text

in between  
the para or any lines

Syntax:

<mark> This is important </mark>

## 14] Bucket List - Naming Reason

14] del and ins tag:

The `<del>` element defines text that has been deleted from a document. Browsers usually strike a line through deleted text.

Syntax:

```
<p> My favorite colour is <del> black  
</del> red </p>
```

The `<ins>` element defines text that has been inserted into a document. Browsers usually insert text.

Syntax:

```
<p> My favorite colour is <del> black  
</del> <ins> blue </ins> </p>
```

15] Sub and Sup:

Sub - Subscript text. It appears half a character below the normal line and sometimes rendered in small font or font size (i.e.  $H_2$ ).

The <sup> element defines superscript text. It appears half a character above the normal line and rendered sometimes in small text. (i.e. www', a<sup>2</sup>)

Syntax:

<p> i like 4<sub>2</sub>0</p>

<p> The power of 2<sup>2</sup> square is 4 </p>

## 16] Quotation Tag :- (Q-Tag)

This tag is used to double quote the lines in html.

Syntax:

<p> i'm currently undergone my <q> kidney transplant previous month 9<sup>th</sup> march 2025 </q> it's been a 2<sup>nd</sup> life for me </p>

## 17] Abbreviation Tag : (abbr tag)

Abbreviate an word when cursor picks over on it.

Syntax:

<hi> Hello i'm <abbr title="Dinesh">DK</abbr>  
</hi>

18] More on Comments:

<!-- This is a Comment -->

<h1> I love <!-- you --> me </h1>

<!DOCTYPE html> <!DTP----->

## Advanced/Intermediate

① Target attribute in link tag:

Opening the link on another tab

<a href = "www.fb.com"> Click </a>

Syntax:

<a href = "www.fb.com" target = "\_blank"> Click  
</a>

It specifies where to open the linked document.

② Image as a link tag

Opening the link by clicking the Png.

<a href = "my-image.png"> Click here </a>

Syntax:

<a href = "https://Shopping-512.png" height = "50px">  
</a>

③ Mailto

This is an attribute which is used to mailing

Syntax:

```
<a href="mailto:dk@gmail.com">email me</a>
```

#### ④ FavIcon

It is a small image displayed next to the page title in the browser tab.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<link rel="icon" type="image/x-icon" href  
      ="images/web-img.jpg">
```

```
</head>
```

```
</html>
```

link → Used to link JS & CSS files.

rel → it is used to define type of the content linked to.

link defines the relationship between a document and an external resource.

---

## 3] Table tag in html

Html tables allow web developers to arrange data into rows and columns. A table in html consist of rows and columns.

Syntax:

<table>

<thead>

<tr>

<th> Sno </th>

<th> hobbies </th>

</tr>

</thead>

<tbody>

<tr>

<td> 1 </td>

<td> Playing cricket </td>

</tr>

<tr>

<td> 2 </td>

<td> Watching Movie </td>

</tr>

</tbody>

</table>

## Adding captions:

Caption tag mentions what kind of table

It is. `<caption>` `</caption>` = Defines tables caption.

`<table>`

`<caption>` Something about me `</caption>`

`<th>` ~~`<th>`~~

`<tr>`

`</tr>`

`<td>`

`</td>`

`<tr>`

`<td>`

`</tr>`

## Table Cells :-

Each table cell is defined by `<td>` and `</td>`

td stands for table data.

Each table row starts with `<tr>` and ends with a `</tr>` tag. tr stands for table row.

table heading name each row. th stands for table header.

## Coloumn Span and Row Span

Spanning over multiple columns Use the Colspan attribute. (spanning across Coloumns).

Spanning over multiple rows Use the Rowspan attribute. (spanning across rows).

Syntax :

<thead>

<tr>

<th colspan="2"> project </th>

<th> lang </th>

</tr>

</thead>

<tbody>

<tr>

<td> URBA </td>

<td rowspan="2"> Java </td>

</tr>

<tr>

<td> Tree cutting detector </td>

</td>

</tbody>

Styles In Table :-

Style Attribute In Table :

table width, border.

Syntax:

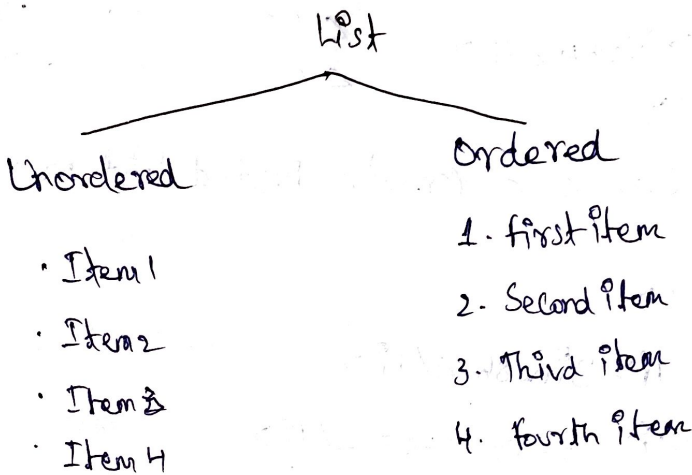
`<table style="width: 300px; border: 1px dashed;">`

~~Use~~ Use dashed for single line of border

~~Use~~ around table.

## ② List tag in html:-

HTML list allows web developers to group a set of related items in lists.



Unordered html list.

An unordered list starts with tag `<ul>` tag. Each list item. The list items will be marked with bullets (small black circles) by default.

Syntax:

You can add styles by using

style attribute on `<ul style="list`

`<ul>`

`style-type=" " ">`

`<li> Coffee </li>`

`<li> Tea </li>`

`<li> Milk </li>`

`</ul>`

Changing shape of the circle

by another shape.

`Style="list-style-type:"`

## Ordered List :-

An ordered list starts with `<ol>` tag.  
Each list item starts with `<li>` tag.

### Syntax :-

`<ol>`

`<li> item1 </li>`

`<li> item2 </li>`

`<li> item3 </li>`

`<li> item4 </li>`

`</ol>`

You can change the numbers  
marked numbers by upper  
lower case letters and the  
roman letters by

`<ol style="type='A'">`

`<ol style="type='a'">`

`<ol style="type='I'">`

`<ol style="type='i'">`

You can also create nested list or sub-list.

`<ol>`

`<li> Dinesh </li>`

`<li> 29-06-2005 </li>`

~~`<li>`~~ `<ul>`

`<li> from Tirupur`

`<li> 19yrs old`

`</ul>`

`</ol>`

## 3) div and Span:

It is an default container in HTML  
div is an block element full width.

The `<div>` element is by default a block  
element, meaning that it takes all available

width and comes with lines break before and after. This has no required attributes but style, class and id are common. The div element is often used to group sections of a web page together. ~~it~~.

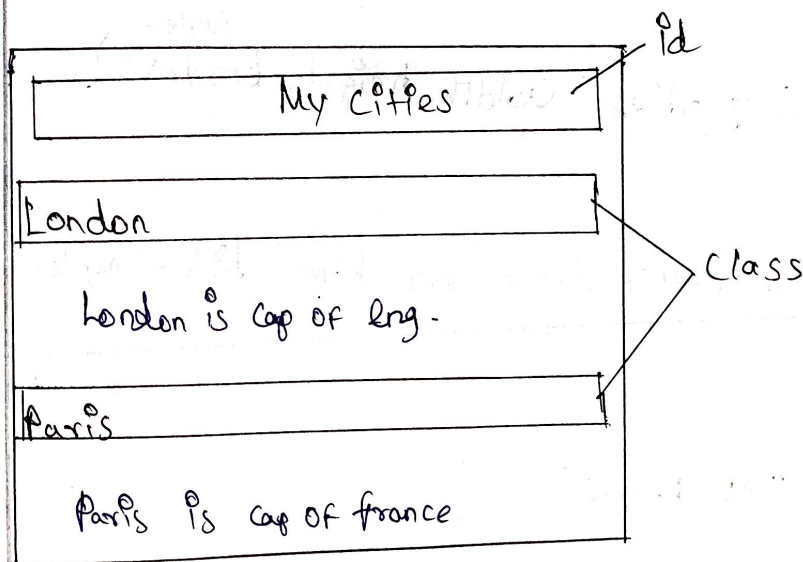
The span is an inline element which means it doesn't start on a new line and only occupies as much spaces as its content requires.

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Class and id :

A Class name can be used by multiple HTML elements, while an id name must only be used by one HTML element within the page.



Syntax for id :-

Id = "img" Navigating something.

Id = "img"

<a href = "img">Image </a>

5] IFrame Tag :-

Accessing or putting another website fully in our own website. It is used to display a web page within a web-page. <iframe> tag specifies an inline frame. An inline frame is used to embed another document within the current HTML document.

<iframe src = "uri" title = "description" </iframe>

Properties : Width, height, <sup>style =</sup> "border".

<iframe src = "learn.html" title = "my learning">  
</iframe>

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6] HTML Forms :-

Forms is used to collect user input. The user input is most often sent to a server for processing.

## <form> element:

The html element <form> is used to create an HTML form for User input.

<form>      The <form> element is a container  
form elements      for different types of input element  
Such as: text field, Check box,  
</form>      Radio button, Submit button, etc...

## <input> element:

It is the Most Used form element. This  
can be displayed in many ways, depending on type  
attribute.      Examples:

<input type="text">      Disp single line text input field  
<"      "      "radio">      Disp radio button (Selecting one  
of many choice)  
<"      "      "checkbox">      Disp checkbox (Selecting 0 (or)  
more of many choice)  
<"      "      "Submit">      Disp a Submit button  
<"      "      "button">      Disp a clickable button

## <label> element:

defines a label for many form elements. It is  
Usefull for Screen-reader Users. bcoz it will read  
out loud the label when the user focuses on input  
element.

The HTML <form> elements:

- .) <input> .) <label> .) <select> .) <textarea>
- .) <button> .) <fieldset> .) <legend> .) <datalist>
- .) <output> .) <option> .) <optgroup>

Description:

- <form> - Defines an HTML form for user input
- <input> - " " Input control
- <textarea> - " " a multiline i/p control (textarea)
- <label> - " " a label for an <i/p> element
- <fieldset> - Groups a related element in form
- <legend> - " " a Caption for a <fieldset> element
- <select> - " " a drop down list
- <optgroup> - " " a grp related option in drop-down list
- <option> - " " in a drop down list
- <button> - " " a Clickable button
- <datalist> - Specifies a list of pre-defined opt for i/p control.
- <output> - " " the result of a calculation.

## HTML Input types:

<input type = "button", "checkbox", "color", "date",  
"datetime-local", "email", "file", "hidden", "image",  
"month", "number", "password", "radio", "range",  
"reset", "search", "submit", "tel", "text", "time",  
"url", "week" >

The default value of the type attribute is text.

## Input restriction:

|                |                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------|
| checked        | Specifies if field should be pre-selected when the page loads (for type = "checkbox" or type = "radio"). |
| disabled       | Specifies that an input field should be disabled.                                                        |
| max            | Specifies maximum value for an input field                                                               |
| maxlength      | Specifies the maximum number of char for <sup>for</sup> <sub>field</sub>                                 |
| min            | Minimum value for an input field                                                                         |
| pattern        | Specifies a regular expression to check for <sup>for</sup> <sub>field</sub>                              |
| readonly       | Specifies that an <sup>for</sup> <sub>field</sub> is read only                                           |
| required       | Input field is required (must be filled out).                                                            |
| size           | Specifies the width (in char) of an <sup>for</sup> <sub>field</sub>                                      |
| step           | Specifies the legal number interval for <sup>for</sup> <sub>field</sub> .                                |
| value          | default value for an input field                                                                         |
| Type attribute | <input type = " " > specifies the input type to display.                                                 |

Example Syntax:

```
<form>
```

```
<label> Name </label>
```

```
<input type="text">
```

```
</form>
```

7] fieldset and legend tag:-

The <fieldset> tag is used to group related elements in a form. It draws a box around the related elements.

The <legend> tag is used to <sup>define a</sup> ~~group related~~ ~~elements~~ caption for the <fieldset> element.

Example Syntax:-

```
<fieldset>
```

```
<form> <legend> Personal Details </legend>
```

```
<form>
```

```
<label> Name </label>
```

```
<input type="text">
```

```
<label> DOB </label>
```

```
<input type="Date" required>
```

```
</form>
```

```
</fieldset>
```

## Semantic tags.

It describes the meaning of the content. not just how it looks.

`<header>` - It specifies a header for a document or section.

`<nav>` - Defines navigation links

`<Section>` - Defines a section in document.

`<footer>` - footer for document or section

`<figcaption>` - Defines a caption for a `<figure>` element.

`<figure>` - Self-Contained container, like illustration, diagrams, photos, code listing, etc...

`<main>` - main content of a document

`<aside>` - Defines content aside from the page doc.

`<mark>` - Marked / highlighted text.

`<article>` - Independent, self-contained content

`<details>` - additional details that the user can view or hide.

`<summary>` - visible heading for a `<details>` element.

`<time>` - Defines a date/time.

1) <header>

<header>

<h1> Website </h1>

<p> Welcome to my site </p>

</header>

2) <footer>

<p> © Copy; 2025 Dinesh. All rights reserved </p>

</footer>

3) <nav>

<ul>

<li><a href="#">home </a> </li>

<li><a href="#contact">Contact </a> </li>

</ul>

</nav>

4) <main>

<h2> Main Content </h2>

<p> This is the primary content </p>

</main>

5) <section>

<h3> latest </h3>

<p> What's Cookin </p>

</section>

6) <article>

<h2> How to learn HTML </h2>

</article>

7) <aside>

<h4> advertisement </h4>

</aside>

8) <figure>

<img src = "nature.jpg">

<figcaption> A peaceful view of nature </figcaption>

</figure>

9) <time>

<p> event starts at <time datetime = "2025-06-01">

9:00 AM, June 1st, 2025 </time> </p>

Semantic Tag → Replaces what?

<header> → replaces <div id = "header"> top part  
like logo/  
title

<footer> → " < " > (bottom part like copyright/link)

<nav> → " < " > (for navigation links)

<main> → " < " > (main content area of page)

<section> → " < " > (grp of related content)

<article> → " < " > (Independent Content like blog post)

<aside> → " < " > (side content like ads or tips)

<figure> → " < " > (for image + caption block)

<figcaption> → " < p class = "caption"> (caption for an img)

<time> → " < span class = "time"> (to show date/time)

~~<header>~~ does not ~~go~~  
Inside `<body>`.  
Semantic tags replace unnecessary ~~tags~~ <sup>div/spans</sup>  
Not actual HTML structure tags like `<body>`,  
`<html>`, etc...

## Meta Tags

`<meta>` tags live inside `<head>` of your HTML.

They provide information about the webpage  
(not visible to users) [data about the data (seo)].

They help search engines, browsers and social  
media platforms understand your page.

Syntax:

```
<meta name=" " content=" " >
```

Common meta tags with examples:-

| Purpose | Syntax | Description |
|---------|--------|-------------|
|---------|--------|-------------|

Charset: ~~meta~~

```
<meta charset="UTF-8">
```

Sets Character encoding (UTF-8) supports most  
languages.

Viewport:

<meta name="viewport" Content="width=device-width, initial-scale=1.0">

It makes site responsive on Mobile

Description:

<meta name="description" Content="This is my web">

helps seo - shows up in Search engine results.

Author:

<meta name="Author" Content="Dinesh Kumar">

Tells who wrote the page.

Keywords:

<meta name="Keyword" Content="Html, web site">

old  
^ Seo method (less used now).

Refresh:

<meta http-equiv="refresh" Content="10">

Auto refresh the every 10 seconds.

Robots:

<meta name="robots" Content="index, follow">

Controls how Search engines crawl the page.

Theme Colour:

<meta name="theme-color" Content="#black">

Sets address bar color in mobile browsers.